

Macro-Economics II

Chapter 1

Equilibrium Income Determination

- **Outline of the chapter**
 - The Components of income
 - Keynesian Cross and the Economy in Equilibrium
 - Fiscal policy and Multipliers
 - Government Purchase multiplier
 - Tax multiplier
 - The Balanced Budget Multiplier

1.1 Keynesian Cross and the Economy in Equilibrium

- According to the Keynesian Theory, equilibrium condition is generally stated in terms of aggregate demand (AD) and aggregate supply (AS). An economy is in equilibrium when aggregate demand for goods and services is equal to aggregate supply during a period of time.

So, equilibrium is achieved when:

- $AD = AS \dots (1)$

We know, AD is the sum total of Consumption (C) and Investment (I):

- $AD = C + I \dots (2)$

Also, AS is the sum total of consumption (C) and saving (S):

- $AS = C + S \dots (3)$

Substituting (2) and (3) in (1), we get:

- $C + S = C + I$

Or, $S = I$

- **Two Approaches for Determination of Equilibrium Level output and employment in the economy are:**
 1. Aggregate Demand-Aggregate Supply Approach (AD-AS Approach)
 2. Saving-Investment Approach (S-I Approach)
- It must be kept in mind that AD, AS, Saving and Investment are all **planned** or **ex- ante** variables.

Assumptions made in determination of equilibrium output:

- (i) The determination of equilibrium output is to be studied in the context of **two-sector model** (households and firms). It means, it is assumed that there is no government and foreign sector.
- (ii) It is assumed that investment expenditure is **autonomous**, i.e. investments are not influenced by level of income.
- (iii) Price level is assumed to **remain constant**.
- (iv) Equilibrium output is to be determined in context of short-run.

Aggregate Demand-Aggregate Supply Approach (AD-AS Approach):

- According to the Keynesian theory, the equilibrium level of income in an economy is determined when aggregate demand, represented by $C + I$ curve is equal to the total output (Aggregate Supply or AS).

Aggregate demand comprises of two components:

1. Consumption expenditure (C):

- It varies directly with the level of income, i.e. consumption rises with increase in income.

$$C = a + bY_d$$

2. Investment expenditure (I):

- It is assumed to be independent of the level of income, i.e. investment expenditure is **autonomous**. So, AD curve is represented by $(C + I)$ curve in the income determination analysis.
- Aggregate supply is the total output of goods and services of the national income. It is depicted by a 45° line. Since the income received is either consumed or saved,

- AS curve is represented by the $(C + S)$ curve.

Employment (Lakhs)	Income (V)	Consumption (C)	Saving (S)	Investment (I)	AD C + I	AS C + S	Remarks
0	0	40	-40	40	80	0	$AD > AS$
10	100	120	-20	40	160	100	$AD > AS$
20	200	200	0	40	240	200	$AD > AS$
30	300	280	20	40	320	300	$AD > AS$
40	400	360	40	40	400	400	Equilibrium ($AD = AS$)
50	500	440	60	40	480	500	$AD < AS$
60	600	520	80	40	560	600	$AD < AS$

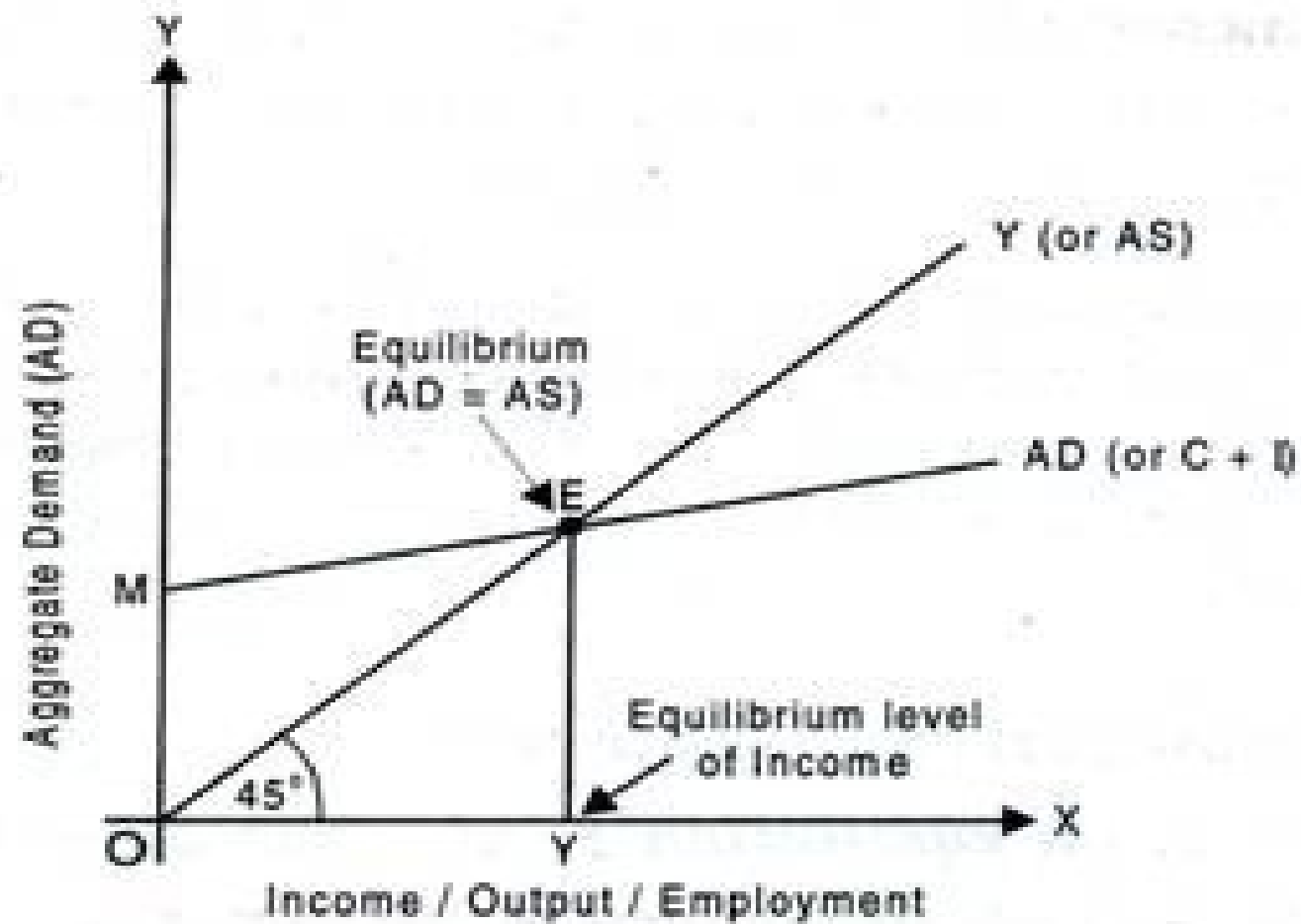


Fig. 8.1

- In the above figure , the AD or (C +I) curve shows the desired level of expenditure by consumers and firms corresponding to each level of output. The economy is in equilibrium at point 'E' where (C + I) curve intersects the 45° line.
- 1. 'E' is the equilibrium point because at this point, the level of desired spending on consumption and investment exactly equals the level of total output.
- 2. OY is the equilibrium level of output corresponding to point E.
- 3. In Table 8.1, the equilibrium level of income is \$400, when AD (or C +I) = AS = \$ 400.
- 4. It is a situation of 'Effective Demand'. Effective demand refers to that level of AD which becomes 'effective' because it is equal to AS.
- If there is any deviation from the equilibrium level of output, i.e. when planned spending (AD) is not equal to planned output (AS), then a process of **readjustment** will start in the economy and the output will tend to adjust up or down until AD and AS are equal again.

- When planned spending (AD) is more than actual output (AS), then $(C + I)$ curve lies above the 45° line. It means that consumers and firms together would be buying more goods than firms are willing to produce. As a result, the planned inventory would fall below the desired level.
- To bring the inventory back to the desired level, firms would resort to increase in employment and output until the economy is back at output level OY, where AD becomes equal to AS and there is no further tendency to change.

- **When AD is less than AS:**
- When $AD < AS$, then $(C + I)$ curve lies below the 45° line. It means that consumers and firms together would be buying less goods than firms are willing to produce. As a result, the actual inventory would rise. To clear the unwanted increase in inventory, firms plan to decrease the employment and output until the economy is back at output level OY , where AD becomes equal to AS and there is no further tendency to change.

- It must be noted that equilibrium level may or may not be at the level of full employment, i.e. equilibrium is possible even at a level lower than the full employment level.

- **Saving-Investment Approach (S-I Approach):**
- According to this approach, the equilibrium level of income is determined at a level, when planned saving (S) is equal to planned investment (I).

- In the figure below, Investment curve (I) is parallel to the X-axis because of the autonomous character of investments. The Saving curve (S) slopes upwards showing that as income rises, saving also rises.
- 1. The economy is in equilibrium at point 'E' where saving and investment curves intersect each other.
- 2. At point 'E', ex-ante saving is equal to ex-ante investment.
- 3. OY is the equilibrium level of output corresponding to point E.
- 4. In table below, the equilibrium level of income is \$400, when planned saving – planned investment = \$ 400
- If there is any deviation from the equilibrium level of income, i.e., if planned saving is not equal to the planned investment, then a process of readjustment will start which will bring the economy back to the equilibrium level.

- **When saving is more than Investment:**
- If planned saving is more than planned investment, i.e. after point 'E' in Fig. 8.2, it means that households are not consuming as much as the firms expected them to. As a result, the **inventory rises** above the desired level. To clear the unwanted increase in inventory, firms would plan to reduce the production till saving and investment become equal to each other.

- **When saving is less than Investment:**
- If planned saving is less than planned investment, i.e. before point 'E' in Fig. 8.2, it means that households are consuming more and saving less than what the firms expected them to. As a result, planned inventory would fall below the desired level. To bring the inventory back to the desired level, firms would plan to increase the production till saving and investment become equal to each other.

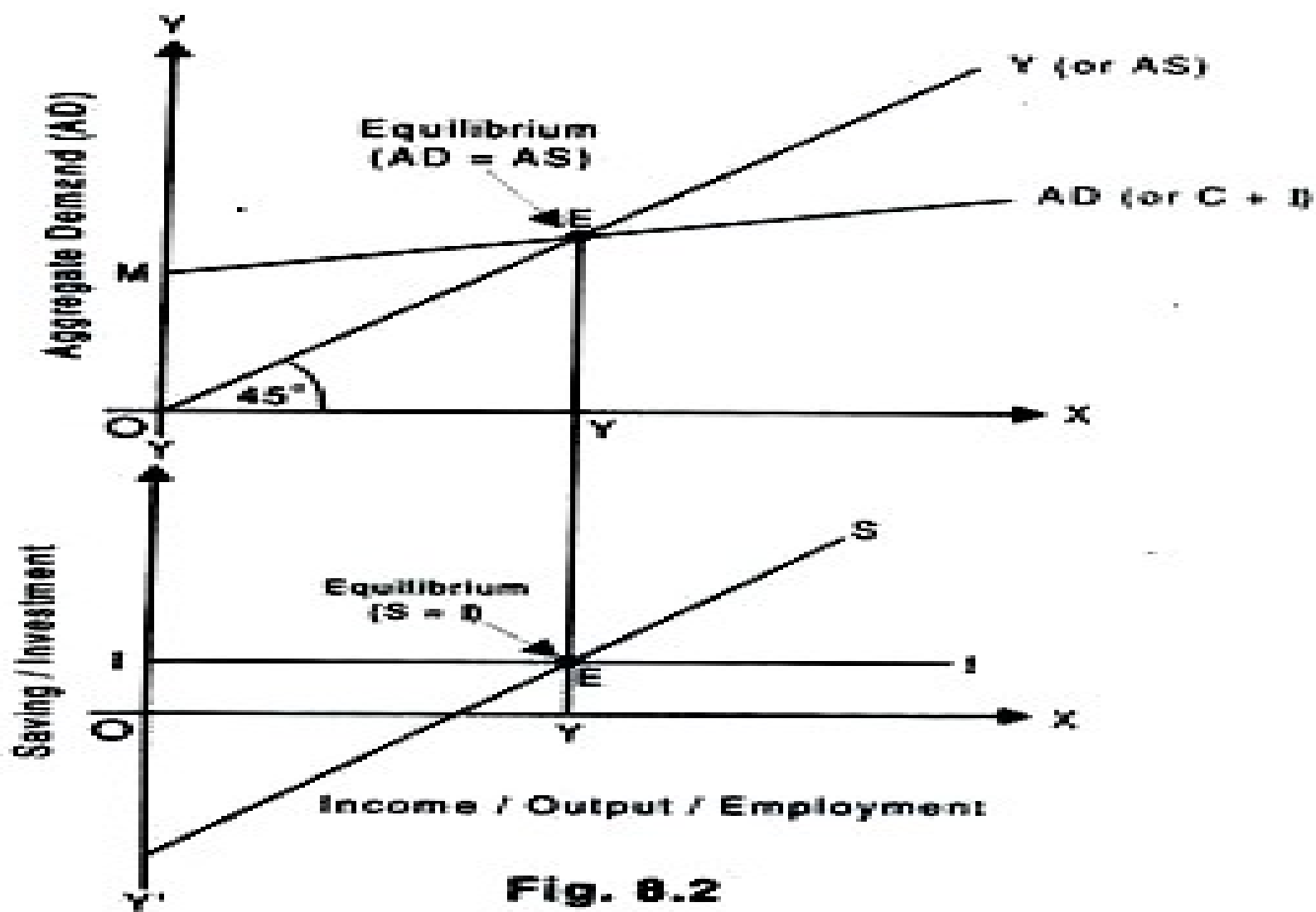


Fig. 8.2

- Fiscal policy and Multipliers
- There are two major government fiscal policy instruments in the commodity market. There are government purchase and government tax revenue.
- Any change in this variables has a multiple effect on real factors such as output.
- The factors which show the relationship between the changes in the instrument and the effect are known as **multipliers**. Thus, the corresponding multipliers are known as government purchase multiplier and tax multiplier
 - Government Purchase multiplier
 - Since government purchases are one significant component of total expenditure, the government purchases imply, for a given level of income, higher planned aggregate demand.
 - If government purchases rise by ΔG , then the planned aggregate demand schedule shifts upward by ΔG as shown in the figure below

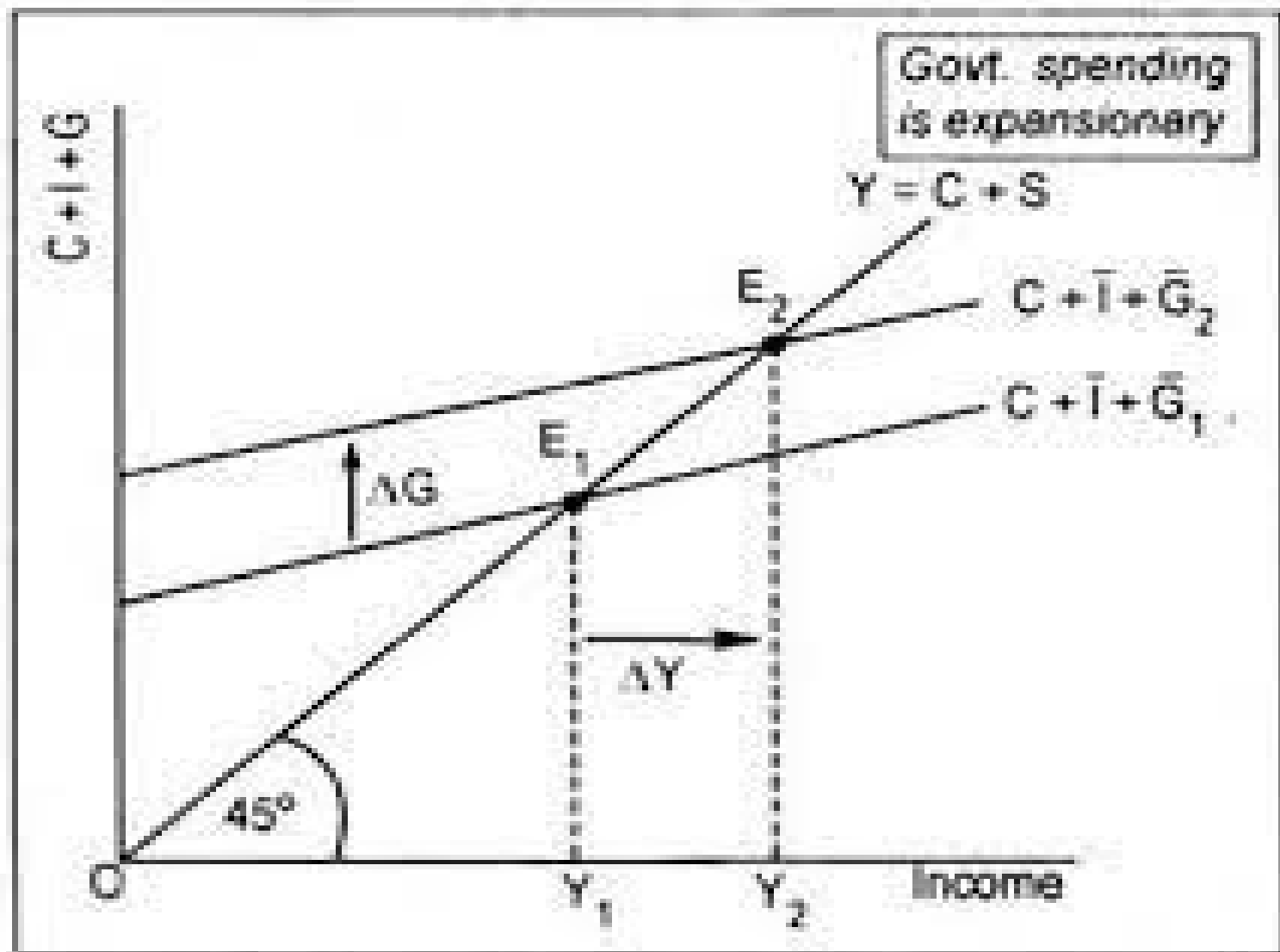


Fig. 3.19: Government Expenditure Multiplier

- The graph shows that an increase in government purchase leads to an even greater increase in income, i.e $\Delta Y > \Delta G$
- As government expenditure increases by ΔG , then the aggregate demand curve shifts upward parallel to itself by the same factor. As a result the equilibrium output increases from Y_1 to Y_2 defined at point E_1 and E_2 respectively.
- It is easy to compare the change in the government spending(ΔG) and the change in output (ΔY)

- We derive multiplier from our national income identity or from aggregate demand
- $Y = a + b(Y - T) + I + G$ assume that T and I are constant and take the total differential of the identity
- $dY = b(dY) + dG$ by simple rearrangement
- $dY - b(dY) = dG$
- $(1 - b)dY = dG$ dividing both sides by $(1 - b)dG$
- We obtain $dY/dG = 1/(1 - b)$ where $b = \text{MPC}$ (marginal propensity to consume)
- We can re write as follows
- $\frac{\Delta y}{\Delta G} = \frac{1}{1 - \text{MPC}}$ or
- $\frac{\Delta y}{\Delta G} = \frac{1}{1 - b}$ then
- $\Delta y = \frac{1}{1 - \text{MPC}} \Delta G$

- Where $b = \frac{dC}{dY} = \text{MPC}$
- The value of MPC lies between 0 and 1
therefore $1/1-b$ is always positive and greater than 1
- Thus a change in government expenditure (ΔG) brings about a large change in output (ΔY)
- $\frac{1}{1-\text{MPC}} \Delta G > \Delta G$ since $\frac{1}{1-\text{MPC}} > 1$

- **Tax multiplier**

The government uses the tax regulation for two purposes

- Major source of government revenue (higher tax rate)
- Motivate producers and consumers (lower tax rate)

- Tax has multiple effect on income
- $Y = a + b(Y - T) + I + G$ assume I and G constant and differentiating the above equation we obtain
- $dY = b(dY - dT) + dI + dG$ since I and G are assumed to be constant $dI = 0, dG = 0$ thus the relationship is simplified as follows
- $dY = b dY - b dT$
- $dY - b dY = -b dT$
- $dY(1 - b) = -b dT$ dividing both sides by $(1 - b)dT$, we obtain the following tax multiplier
- $\frac{dY}{dT} = \frac{-b}{1 - b}$
- $\Delta Y = \frac{-b}{1 - b} * \Delta T$

- Since $b = \text{MPC}$ and its value is between zero and one $0 < b < 1$ the sign of the multiplier
- $dY/dT = -b/(1-b)$ in absolute terms the value of tax multiplier is greater than 1.
- Thus the multiplier implies that taxes and income are inversely related and a unit increase in tax leads to a more than proportionate decrease in income.

- For instance an increased tax by ΔT immediately reduces disposable income ($Y-T$) by ΔT . therefore, consumption also decreases by $(MPC) \Delta T$. for any level of income Y , aggregate demand is now lower.

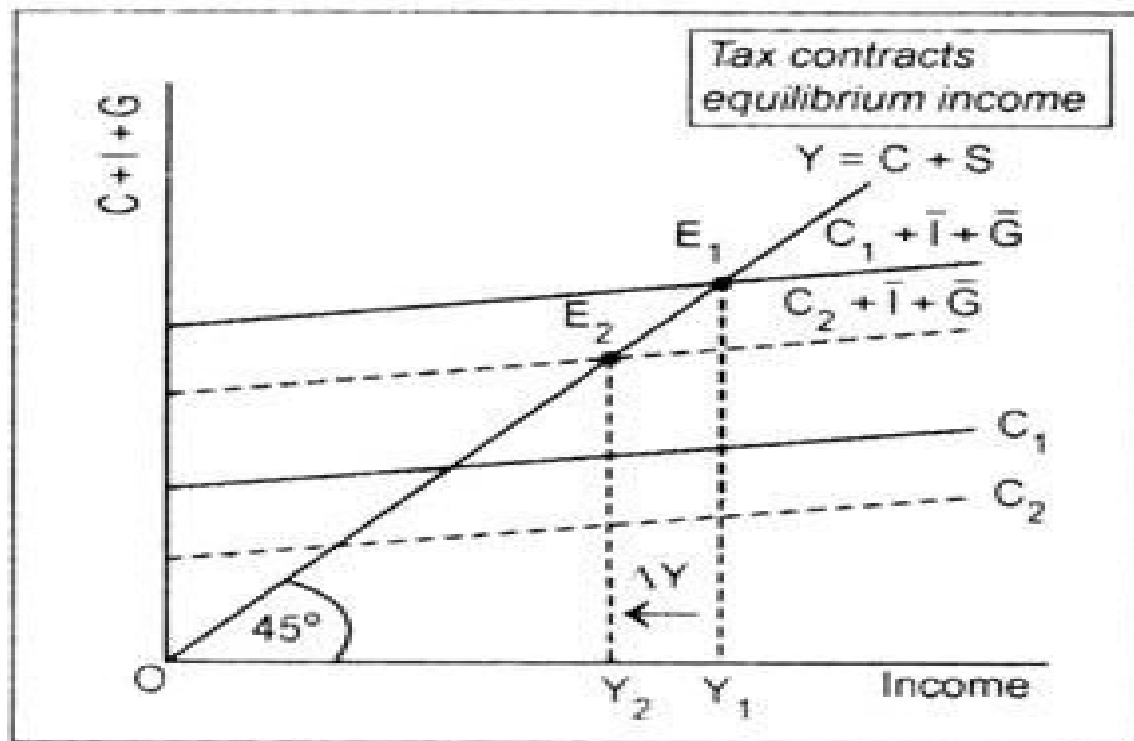


Fig. 3.18: Tax and Equilibrium Income

Exercise

- Given consumption function investment (I), and government expenditure (G) as follows (the values are in millions birr)

$$C=90+0.75Y_d \quad I=120 \quad \text{and} \quad G=180 \quad \text{if} \quad Y_d=Y-T$$

- Calculate the equilibrium level of output and consumption if the tax is equal to $T=80$
- Calculate government purchase multiplier and interpret the result
- Calculate government tax multiplier and interpret the result
- Calculate the change in output and the new equilibrium output if government expenditure increased by 80 units.
- Calculate the change in output and the new equilibrium output if government increases tax by 20 units.

The Balanced Budget Multiplier

- Balanced budget multiplier is to find the total impact of a simultaneous change in government spending and tax.
- Different combination of the elements have different results in the change of output.
- Lets see the impact of a combination of the two where an increase in government expenditure is exactly equal to the change in tax. This may be the case when government collects more tax to increase its spending as part of its macroeconomic policy.

- We know that government spending increases the aggregate demand and taxes reduces the aggregate demand. If government is planning to increase spending by 100million birr at the same time government want to increase the tax collection by 100 million birr, the change in income is one (the multiplier is one). This is called balanced budget multiplier.
- The reason is that the impact of government expenditure on output is **partly negatively** offset by the negative impact of tax.
- For instance, if both government expenditure and the tax are increased by 50 units, then the national output will be increased by 50 units.

Mathematically to prove that balanced budget multiplier is one.

- To determine the resulting change in income, we add the change in income to the change in government purchase or spending $\Delta Y = \frac{1}{1-b} \Delta G$ and the change in income due to a change in taxes.
- $\Delta Y = \frac{-b}{1-b} \Delta T$ thus, the total change can be written as follows
- $\Delta Y = \frac{1}{1-b} \Delta G + \frac{-b}{1-b} \Delta T$ since change in government expenditure and change in tax is the same ($\Delta G = \Delta T$), we can substitute ΔT by ΔG and rewrite the expression as follows.
- $\Delta Y = \frac{1}{1-b} \Delta G + \frac{-b}{1-b} \Delta G$
- $\Delta Y = \left(\frac{1}{1-b} - \frac{b}{1-b} \right) \Delta G$
- $\Delta Y = \frac{1-b}{1-b} \Delta G$
- $\Delta Y = \Delta G$

Assignment 1

- Why does fiscal policy have a multiple effect on income?
- What are the criticism of government multiplier?

Chapter 2

Consumption Spending

- **Outline of the chapter**
 - Definition and Concepts of Consumption
 - Theories of Consumption
 - Keynesian absolute income hypothesis
 - Relative Income hypothesis
 - Fisher's Intertemporal Model
 - Modigliani's Life-Cycle Hypothesis
 - Friedman's Permanent Income hypothesis

Definition and Concepts of Consumption

- **Consumption** is defined as the use of goods and services by a household.
- It is a component in the calculation of the [Gross Domestic Product \(GDP\)](#).
- The economy can be taken as the combination of several sectors. These sectors can be household sector, business sector, government sector and foreign sector. All of these sectors demand goods and services from the economy.
- The aggregate demand of all of them is known as aggregate demand and includes consumption of household sector(C), investment spending of business sector (I), government spending (G) and net of exports (NX) which is the difference between export and import (X-M).
- We can represent aggregate demand in the following equation

$$\text{Aggregate demand (AD)} = C + I + G + NX$$

- Macroeconomists are always concerned with **equilibrium level of output** which is that level of output at **which quantity of output supplied is equal to the quantity demanded**.
- As consumption (or more correctly, consumers expenditure) is **the largest component of desired aggregate expenditure**, its determinants of factors affecting the consumption are also the most important determinants of aggregate demand and so general equilibrium.

- Because of this, macroeconomists are highly concerned with the task of explaining the determinant of consumer spending via a well specified consumption functions.
- Macro economists usually want to know or explore factors that determine the consumption of households.
- Therefore Macroeconomists typically use **consumption** as a proxy of the overall economy.
- **Consumption function:-** explains the relationship between consumption and income.
- The keynesian consumption function states that when income increases, consumption also increases **linearly**.
- This hypothesis is also known as **absolute income hypothesis**. It can also be stated that the current real consumer spending is a function of current real disposable income. The simple linear relationship is typically assumes the following relations.
- $C = a + cY_d$; $a > 0$, $0 < c < 1$
- Where C= total consumption of households
 a= autonomous consumption of households
 Y_d disposable personal income
 b= Marginal propensity to consume(MPC)

Marginal Propensity to Consume (MPC)

- It is defined as the proportion of an aggregate raise in pay that a consumer spends on the consumption of goods and services, as opposed to saving it.
- Marginal propensity to consume (MPC) refers to the proportion of extra income that a person spends instead of saves.
- Marginal propensity to consume (MPC) measures how much more individuals will spend for every additional dollar of income.
- MPC is calculated as the ratio of marginal consumption to marginal income.
- MPC is related to the so-called **Keynesian multiplier**, where MPC can help predict the economic growth from a government stimulus.
- The multiplier effect refers to a chain reaction of consumption by various entities brought about by an initial increase in income.
- A MPC of **one** means a person spent all additional income.
- A MPC of **zero** means they spent none of it.

Average propensity to consume

- The average propensity to consume can be referred to as the percentage of income spent on goods and services by an individual.
- It is arrived at by dividing the total amount spent on household consumption by the total disposable income.
- An increase in the average propensity to consume denotes a **high demand** for goods and services.
- An increase or decrease in the average propensity to consume also determines the **propensity to save**.
- The opposite of the average propensity to consume is the average propensity to save.

- The ratio of total saving to total income is called APS. Alternatively, it is that part of total income which is saved. $APS = S/Y$
- For instance, in the following table when national income is 200 billion birr, saving is 30 billion birr. In this case $APS = 30/200 = 0.15$ or 15%.
- If disposable income is 1,000 birr and consumption expenditure is 750 birr, find out APS.
- Ans. $APS = S/Y = 250 (=1,000 - 750)/ 1,000 = \frac{1}{4} = 0.25$ or 25%

- **Marginal propensity to save (MPS):**
- MPS is the ratio of change in saving (AS) to change in income (AY). It is that part of additional income which is saved. In other words, it is a measure of additional saving as proportion of additional (incremental) income. MPS is worked out by dividing change in saving (AS) with the corresponding change in income (AY).
- **Symbolically:**
- $MPS = \Delta S / \Delta Y$
- For instance, in the following table, when national income goes up from Rs 100 crore to Rs 200 crore, saving also goes up from zero to Rs 30 crore.
- In this case, $MPS = \Delta S / \Delta Y = 30 / 100 = 0.3$ or 30%
- **Value of MPS lies always between 0 and 1:**
- Its reason is if additional income is entirely consumed, then there is no saving making $MPS = 0$. If entire additional income is saved, then $MPS = 1$ In short, $0 < MPS < 1$.

- There are different theories about consumption function
 - 1. Keynesian absolute income hypothesis
 - 2. Relative Income hypothesis
 - 3. Fisher's Intertemporal Model
 - 4. Modigliani's Life-Cycle Hypothesis
 - 5. Friedman's Permanent Income hypothesis

Keynesian absolute income hypothesis

- Keynes proposed that consumption is a function of absolute level of income
- $C=f(Y)$ it can be stated as $C=a+bY$ where a is autonomous consumption and b is marginal propensity to consume.
- According to Keynes when income rises, consumption also rises but not as much as the income. The reason why consumption rises less than income is that a part of the increment in income is saved.

- Keynesian consumption function has been depicted by CC' curve in Fig. 6.1 in which along the X-axis national income is measured and along the Y-axis the amount of consumption is measured. In this figure, a line OZ making 45° angle with the X-axis, has been drawn. Because line OZ makes 45° angle with the X-axis every point on it is equidistant from both the X-axis and Y-axis.
- Therefore, if consumption function curve coincides with 45° line OZ it would imply that the amount of consumption is equal to the income at every level of income. In this case, with the increase in income, consumption would also increase by the same amount.

- As has been said above, in actual practice consumption increases less than the increase in income. Therefore, in actual practice the curve depicting the consumption function will deviate from the 45° line. If we represent the above consumption schedule by a curve, we would get the propensity to consume curve such as CC in Fig. 6.1.
- It is evident from this figure that the consumption function curve CC' deviates from the 45° line OZ. At lower levels of income, the consumption function curve CC lies above the OZ line, signifying that at these lower levels of income consumption is greater than the income.
- It is so because at lower levels of income, a nation may draw upon its accumulated savings to maintain its consumption standard or it may borrow from others. As income increases, consumption also increases and at the income level OY_0 , consumption is equal to income.

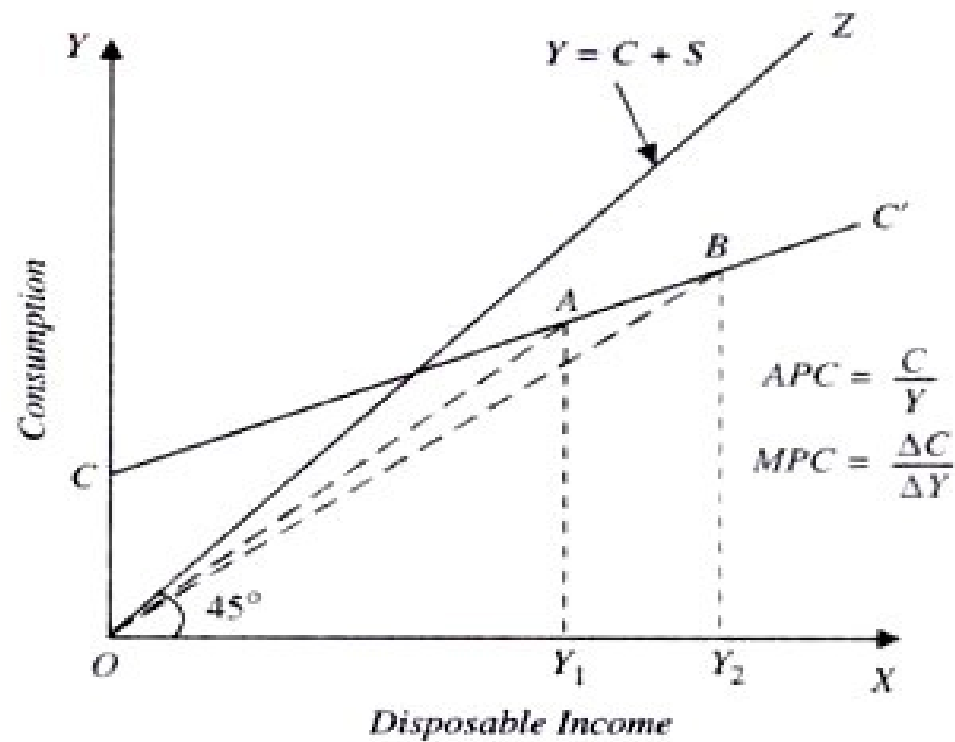


Fig. 6.3. *Consumption Function: Declining Average Propensity to Consume*

- Beyond this, with the increase in income, consumption increases but less than the increase in income and therefore, consumption function curve CC lies below the 45° line OZ beyond Y_0 . An important point to be noted here is that beyond the level of income OY_0 , the gap between consumption and income is widening. The difference between consumption and income represents savings. Therefore, with the increase in income, saving gap also widens and as we shall see later, this has a significant implication in macroeconomics.

- It is useful to point out here that when the consumption function of a community changes, the whole consumption function curve changes or shifts. When propensity to consume increases, it means that at various levels of income more is consumed than before.
- on the contrary, when the propensity to consume of the community decreases, the whole consumption function curve shifts downward signifying that at various levels of income, less is consumed than before.

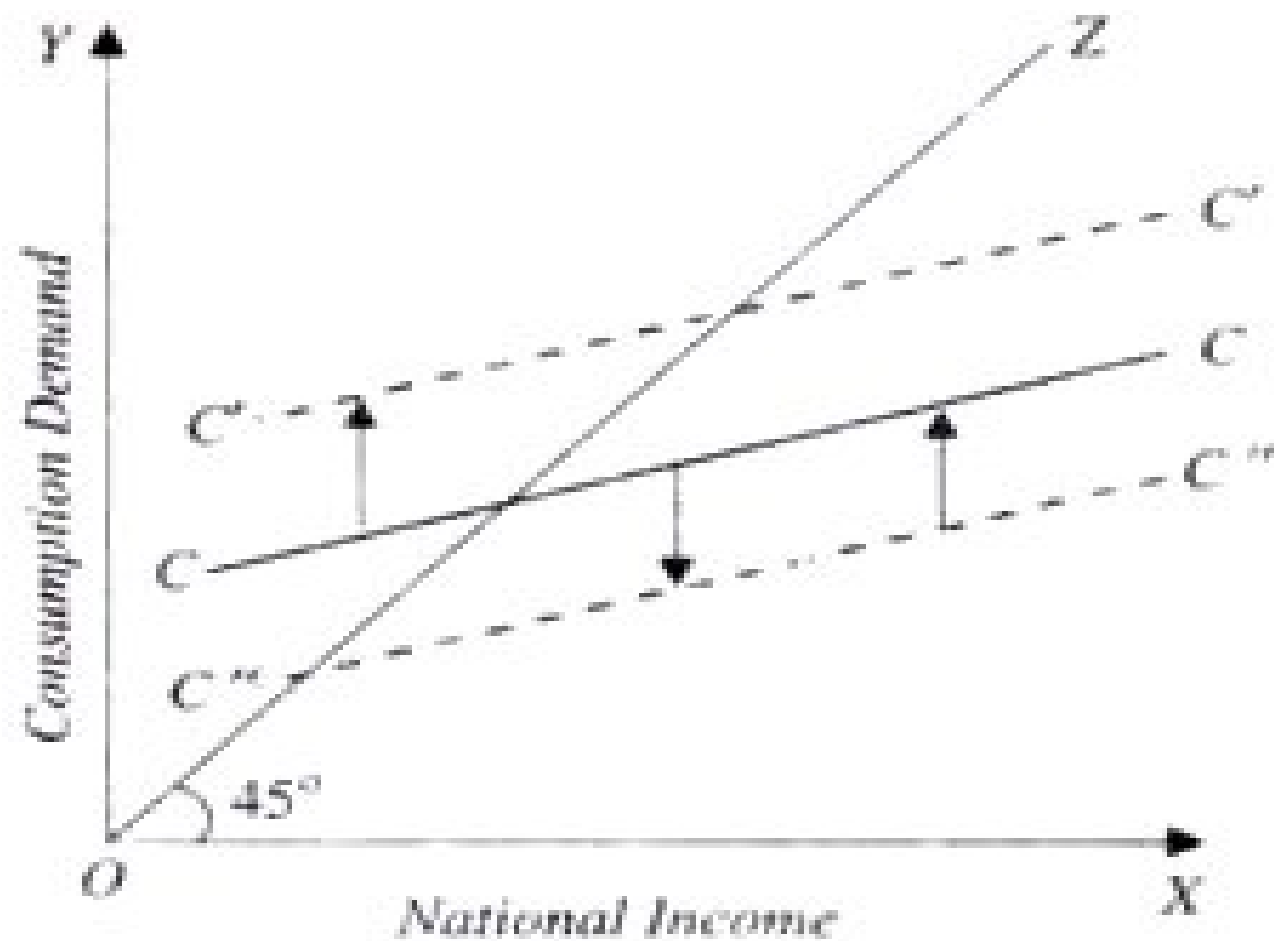


Fig. 6.2. *Shift in Consumption Function*

☞ Therefore the *three* complementary propositions/
assumption of Keynes *can be summarized as*

☞ *1) Consumption will rise as disposable income rises* => *income* is
the most important determinant of consumption

☞ The *higher the income* of a household the *higher will be
consumption*.

☞ *2) The MPC is positive and less than 1* => *the increase in
consumption* is *smaller* than *the increase in disposable income*.

☞ Households *spend less* than their total income

☞ *3) APC is greater than the MPC but assumed to fall as income
rises.*

☞ The implication is that on average *the rich spend less* of their
income (or saves more than) than the poor.

2. Relative Income hypothesis

- Under relative income hypothesis consumption is a function of current income relative to the highest level of income previously attained.
- Income consumption relations are irreversible
- James S. Duesenberry explained as he says that there is a strong tendency in our society for the people to emulate their neighbors and to strive towards a higher standard of living. Based on this he explained an interesting concept called **demonstration effect**.

- This tendency of a family to imitate the consumption habits of rich neighbors or ‘ joneses’ is called demonstration effect.
- It is also known as Duesenberry effect
 - It shows consumption functions are inter dependent
 - Through Demonstration effect, Duesenberry explains the social character of consumption pattern.
- High level of consumption is irreversible
 - Ratchet effect means that households will not reduce their consumption much, as their income falls
 - This is partly due to demonstration effect
 - This is also partly due to the fact that they become accustomed to their previous higher level of consumption, and it is quite hard and difficult to reduce their consumption expenditure when their income has fallen
 - They maintain the earlier consumption level by reducing savings

- Features of Relative Income Hypothesis
 - Consumption is proportional to relative income
 - If relative income falls, APC rises
 - If relative income rises, APC falls
 - In the long run, $APC = MPC$

Relative income hypothesis states that as income increases consumption function curve shifts above so that APC remains constant

3.Fisher"s Inter temporal Model

- Irving Fisher argues that current consumption depends on *lifetime income*.
- According to him, time of income is irrelevant as the consumer can borrow or lend between periods. On the basis of this argument, Irving Fisher developed a model to analyze how rational, forward-looking consumers make consumption choices over a period of time.
- Fisher"s model takes two important assumptions as:
 1. Consumer is forward-looking and chooses consumption for the present and future to maximize lifetime satisfaction.
 2. Consumer"s choices are subject to an inter temporal budget constraint-a measure of the total resources available for present and future consumption

- Given the above assumptions, Fisher's model of inter temporal choice illustrates the following three things:
 - (1) The budget constraints faced by consumers.
 - (2) The preferences between current and future consumption.
 - (3) How constraints and preferences conjointly determine consumer's decision regarding optimal consumption and saving over an extended period of time.

- Rational individuals always prefer to increase the quantity or quality of the goods and services they consume.
- However, most people cannot consume as much as they like due to limited income called budget constraint.
- For the sake of simplicity let us assume that our representative consumer lives for two periods:
 - (a) Period-1 represents consumer's youth life
 - (b) Period-2 represents consumer's old age.Consumer's income and consumption in the two periods are Y_1 , and C_1 and Y_2 and C_2 , respectively.

- In the first period, saving (S) is the difference between income and consumption which is expressed as:

$$S = Y_1 - C_1 \dots (1)$$

In the second period consumption equals the accumulated saving (which includes the interest(r) earned on that saving), plus second-period income which is expressed as:

$$C_2 = (1 + r)S + Y_2 \dots (2)$$

- Where; r = real interest rate (i.e., nominal interest adjusted for inflation).
- Since we do not consider the third period, the consumer is not required to save in the second period

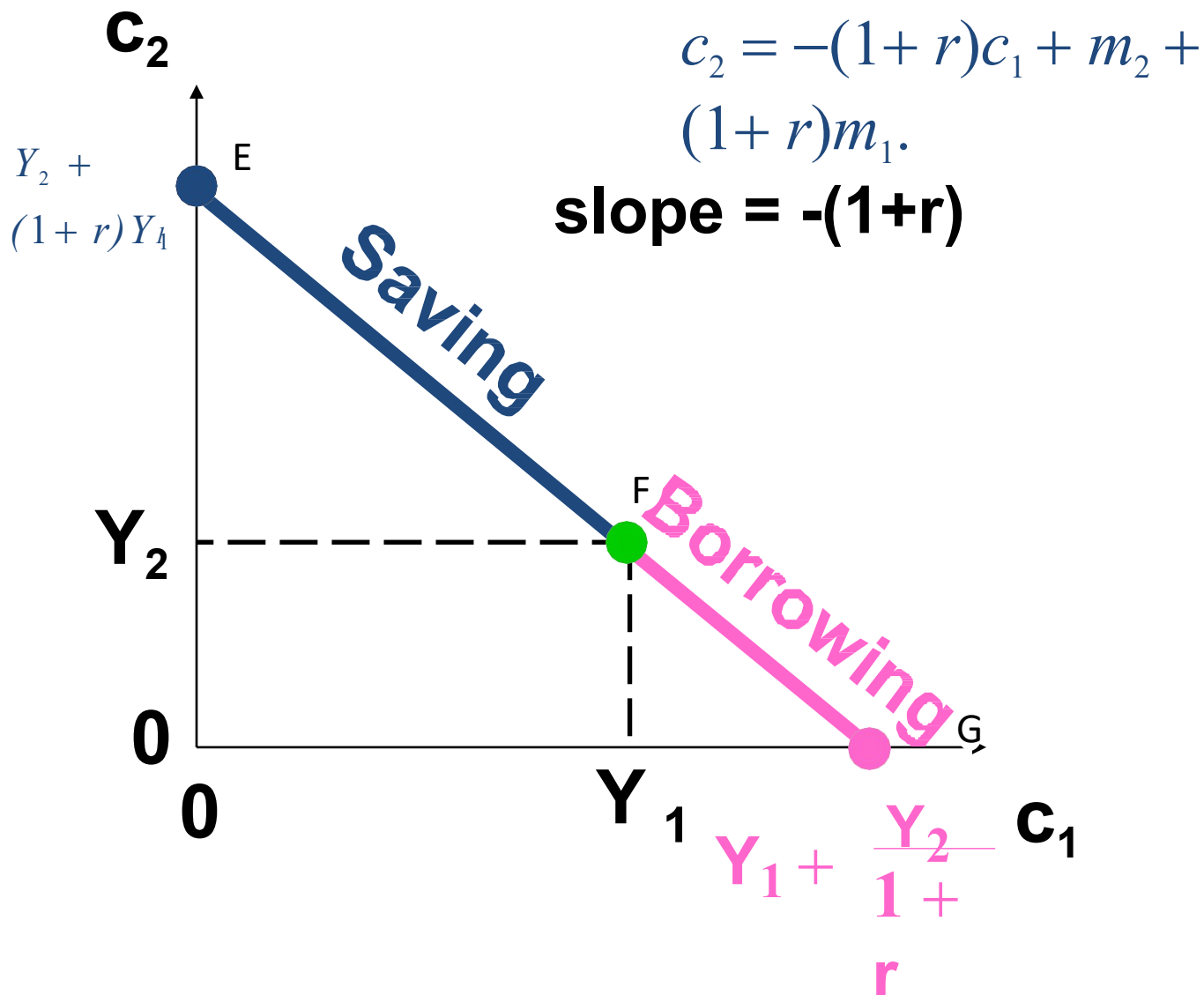
- We can now derive the consumer's budget constraint by combining equations (1) and (2). If we substitute the first equation for S into the second equation we get

$$C_2 = (1 + r)(Y_1 - C_1) + Y_2$$
or, $(1 + r)C_1 + C_2 = (1 + r)Y_1 + Y_2$ [By rearranging the terms]..... (3)
By dividing both sides of equation (3) by $1 + r$ we get

- $$\frac{C_1}{1 + r} + \frac{C_2}{1 + r} = Y_1 + \frac{Y_2}{1 + r} \dots \dots \dots (4)$$

- Since equation (4) relates consumption in two periods to income in both the periods, it expresses the consumers' standard inter-temporal budget constraint.
- As shown in Equation (4),
 - ✓ if $r = 0$, then $C_1 + C_2 = Y_1 + Y_2$, i.e., total consumption in the two periods equals total income in the two periods.
 - ✓ But if $r > 0$, the future consumption- C_2 and future income- Y_2 are to be discounted by a factor $1/(1 + r)$ as $1/(1 + r)$. The discount factor $1/(1 + r)$ measures how much period-1 consumption has to be sacrificed in order to consume 1 unit in period-2.
- Consumer's temporal budget constraint can be shown graphically as below

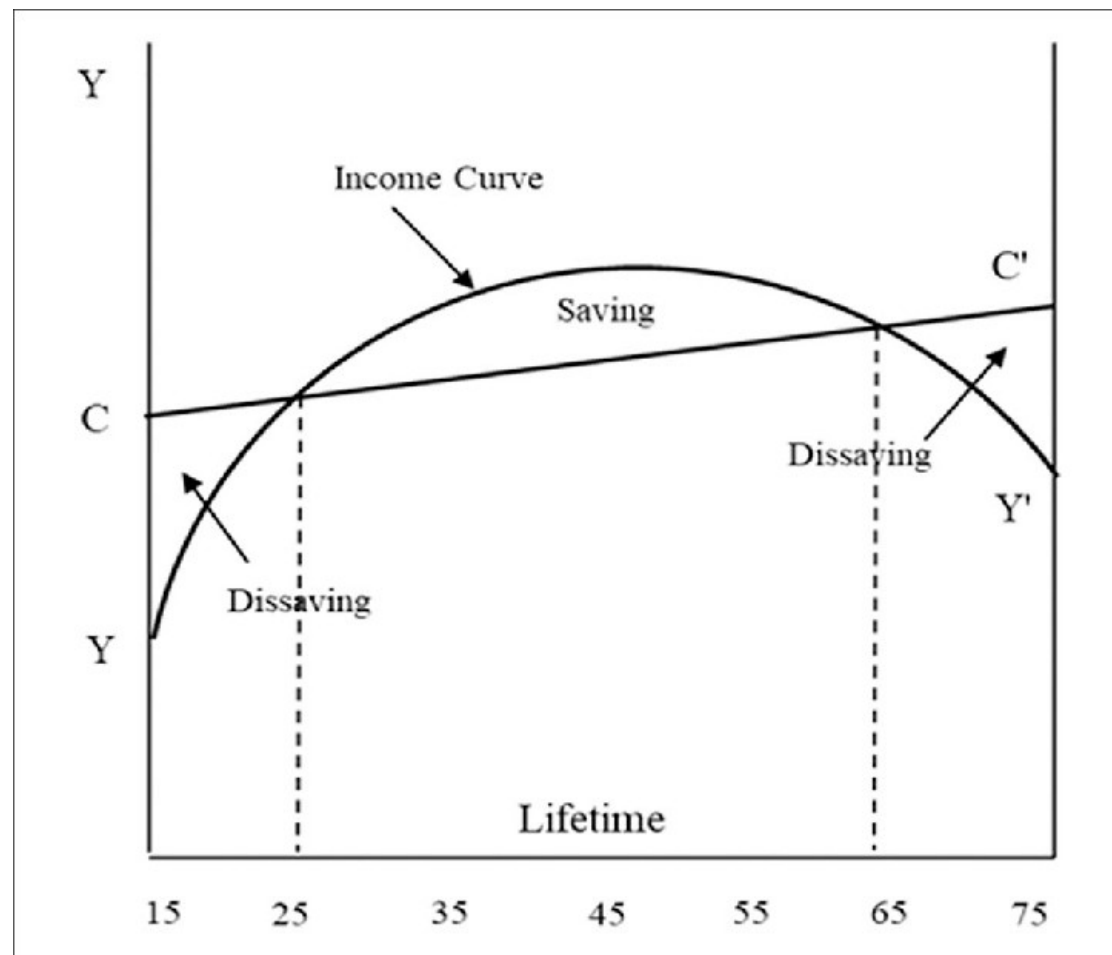
The Intertemporal Budget Constraint



- The above figure shows the alternative combinations of period-1 and period-2 consumption the consumer can choose. It is clear from the above figure that:
 - 1) If the consumer is at point F, he consumes his entire income in both the periods ($Y_1 = C_1$ and $Y_2 = C_2$, Saving(S) = 0, Borrowing (B) = 0)
 - 2) At point E, $C_1 = 0$ and $Y_1 = S$. Therefore $C_2 = (1 + r)Y_1 + Y_2$. Thus if he chooses points between E and F, he consumes less than his income in period-1 and saves the rest for period-2.
 - 3) At point G, $C_2 = 0$. This means that the consumer borrows the maximum possible amount against Y_2 . This means that C_1 is $Y_1 + Y_2/(1 + r)$. Thus if he chooses any point between F and G, he consumes more than his income in period-1 and borrows to make-up the difference.
 - 4) Various other points on the budget line EFG are attainable points.

4. Modigliani's Life-Cycle Hypothesis

- Franco Modigliani tries to give justification for the Keynesians idea average propensity to consume declining function of income and Marginal propensity to consume is less than average propensity to consume and Kuznets idea that average propensity to consume is constant.
- Modigliani supports both seemingly conflicting results about trend of APC the former is true in the short run and the later in the long run.
- According to the model, an individual is a net borrower in earlier years of his/her life and saves during the middle age to repay young age loan and to provide for old age consumption.
- In the model consumption is linear while income is non linear over time. Because individuals have no income or have very low income during their young age and during old age people need recreation which means the person has to work lower hours and as a result receive lower income or they may eventually retire from their job. This situation implies that they may get very low income.



- According to life cycle hypothesis, cross sectional data shows that higher income groups are the middle age group. During middle age there is higher saving relative to consumption which means that there is lower average propensity to consume.
- Furthermore the hypothesis explicitly added wealth as explanatory variable of consumption function.

$$C = \frac{W + RY}{T}.$$

Where

C= consumption

W = Wealth

R = Years until retirement. Remaining years of work

Y = Income

T= Remaining years of life

5. Friedman's Permanent Income hypothesis

- Milton Friedman (1957) According to Friedman, consumption is determined by long-term expected average income rather than current income
- It is the long-term expected income which Friedman calls Permanent Income
- Permanent income may be earned from human and non-human wealth
- Human wealth- labour income
- Non- human wealth consist of physical and financial wealth

- An individual prefers a smooth consumption flow per day rather than plenty of consumption today and little consumption tomorrow
- Thus, people plan their consumption on the basis of expected average income over a long period which he calls as Permanent Income
 - **$C_P = kY_P$**
- ✓ k is the proportion of permanent income consumed and depends on: rate of interest (i), the proportion of non-human wealth to human wealth (w), the desire to add to one's own wealth (u)
 - $k = f(i, w, u)$

- **Permanent and Transitory Income**

- Measured income is the actual income that a family receives during short period
- Measured income consists of: permanent income and transitory income
- A transitory income is a temporary income that is not going to persist in the future
- Measured income may be less than or greater than permanent income depending on transitory income
- $Y_m = Y_p + Y_t$
- Y_t positive implies $Y_m > Y_p$
- Y_t negative implies $Y_m < Y_p$

- According to Friedman, Permanent income is equal to last year's income plus a proportion of change in income occurred between the last year and the current year
 - $Y_p = Y_{t-1} + a(Y_t - Y_{t-1})$
 - Thus, permanent income hypothesis is based on adaptive looking or backward looking expectation.
- There are three major assumptions used by the economists who developed the theory. These are
 1. permanent income and transitory income are uncorrelated meaning there is no systematic or functional relationship between the two variables.
 2. There is no systematic relationship between permanent consumption and transitory consumption
 3. There is no systemic relationship between transitory income and transitory consumption.

Assignment

- **Criticize the five theories of consumption**